

Git Commands

- Log
 - Shows the commit history of the current branch.
 - It includes commit hashes, author, date and messages.
 - *git log*
 - Example output:

```
commit 3a5f7e9d0d1a...
Author: Maham Ismail
Date: Mon Jul 7 14:00:00 2025 +0500

    Added user authentication logic
```
- Remote
 - Manages connections to remote repositories
 - View url of the remote repository
 - *git remote -v*
 - Add a remote github repository by a nickname (origin)
 - *git remote add origin https://github.com/username/repo-name*
 - Remove a remote repository
 - *git remote remove origin*
 - We can use this connection to push,pull and collaborate on github.
- Checkout
 - Switch branches or restore files from a commit.
 - Switch
 - *git checkout main*
 - Restore a file to the last committed version.
 - *git checkout HEAD - - filename.txt*
 - *git restore filename.txt* is also used for the same purpose.
- Revert
 - Creates a new commit that undoes the changes made by a specific commit, it doesn't remove history.
 - *git revert 3a5fe9* (commit id)
- Add
 - Stages changes to be committed.
 - *git add index.html*
- Commit
 - Creates a snapshot of the staged changes.
 - *git commit -m 'new commit'*
- Diff
 - Shows the difference between versions of your files.
 - See unstaged changes
 - *git diff*
 - See staged changes

- *git diff - - cached*
 - Compare branches
 - *git diff main feature-branch*
- Fetch
 - Download changes from a remote repository without merging them.
 - Gets the latest changes from the origin remote but doesn't update your local branch yet.
 - *git fetch origin*
- Push
 - Sends your local commits to a remote repository
 - *git push origin main* (pushes the main branch to the origin remote repository)
- Pull
 - Fetches and automatically merges the changes from the remote repository into your current branch.
 - *git pull origin main*